

# MOUNIKA UDATHALA

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Data Scientist

## SUMMARY

Computer Science graduate specializing in Artificial Intelligence and Machine Learning with hands on experience building predictive analytics systems, NLP applications, intelligent automation workflows, and scalable backend integrations using Python, Scikit learn, SQL, FastAPI, Docker, and cloud technologies. Experienced in developing end to end machine learning pipelines involving data preprocessing, model training, REST API integration, predictive analysis, and AI experimentation for real world applications. Built projects involving NLP based text classification, intelligent prediction systems, vector search workflows, and cloud hosted ML applications while collaborating in agile development environments using Git and modern deployment practices. Strong interest in data science, statistical analysis, AI driven business solutions, and scalable machine learning systems with practical exposure to cloud platforms, data visualization, and intelligent automation technologies.

## SKILLS

**Programming Languages:** Python, Java, JavaScript

**Web Development & Databases:** HTML, CSS, Django, Bootstrap, SQL, MySQL, PostgreSQL, Flask, APIs, REST APIs, FastAPI

**AI & Machine Learning:** Artificial intelligence, Machine Learning, Deep Learning, Data Science, Data Analytics, Natural Language Processing (NLP), Scikit learn, Pandas, NumPy, AI, Machine learning Algorithms, Neural Networks, Reinforcement Learning, Pattern Recognition, SVM, XGBoost, LLMs, RAG, PyTorch, LangChain, LangGraph, OpenAI, CNNs, RNNs, TensorFlow

**Cloud & MLOps:** AWS, Docker, CI/CD Pipelines

**Data Visualization:** Matplotlib, Seaborn, Streamlit, Data preprocessing

**Development Tools & Version Control:** Git, VS Code, Jupyter Notebook, Unit Testing

## WORK EXPERIENCE

### CYPWNG Software Technologies

AI Developer Intern

Mar 2025 - Jun 2025

- Architected RESTful APIs with Django REST Framework and Python, interfacing MySQL and PostgreSQL to serve 10,000 patient records; API design enabled secure EMR integration, cut data retrieval latency by 30% and ensured meticulous SQL query validation for data integrity.
- Engineered AI driven diagnosis engine in Python using scikit learn and TensorFlow (deep learning), trained on 5,000 anonymized cases; model achieved 85% accuracy, automated triage reduced manual review time by 40% while maintaining strict attention to detail in data handling and security.
- Designed NLP pipeline with spaCy to extract symptoms from free text clinical notes, transformed text into feature vectors for machine learning models; improved symptom capture rate by 25%, demonstrating analytical problem solving and seamless integration into the UI.
- Built data analytics dashboards in Python (Pandas, Matplotlib, Streamlit) presenting key health metrics to 200+ daily users; automated reporting saved 15 hours/week, communicated insights to clinicians, and reinforced precision through rigorous unit testing and version control.
- Worked with structured healthcare datasets using Python, Pandas, SQL, and Scikit learn to support predictive analytics workflows, data preprocessing, and intelligent automation systems.
- Assisted in analyzing operational data trends and optimizing REST based backend workflows through data driven reporting and scalable API integrations using FastAPI and PostgreSQL.

## PROJECTS

### Intelligent Air Quality Prediction and Health Advisory System | [github.com/mounika103-goud/air-quality-advisory.git](https://github.com/mounika103-goud/air-quality-advisory.git)

- Developed a Streamlit based machine learning web application that predicts AQI levels using real time environmental data collected via API integration, processed with Pandas and NumPy, and modeled using Scikit learn for accurate air quality forecasting.
- Designed an interactive dashboard with Matplotlib and Seaborn visualizations and implemented personalized health advisory messages based on predicted AQI levels to help users understand air conditions, health risks, and take preventive actions.

### Fake News Detection System using NLP and Machine Learning

- Built a supervised NLP based text classification system using Logistic Regression, Random Forest, and Naïve Bayes to automatically detect and flag misleading or false news articles based on linguistic patterns and content structure
- Developed an end to end Python pipeline including text cleaning, tokenization, TF IDF vectorization, model training, evaluation, and integration of credibility indicators to ensure scalable, accurate predictions and reduce misinformation spread.

### Phishing Website Detection using Machine Learning | [github.com/mounika103-goud/phishing.git](https://github.com/mounika103-goud/phishing.git)

- Developed a phishing detection system using Python and Scikit learn to classify malicious URLs.
- Extracted URL, domain, and webpage based features for predictive analysis.
- Trained ML models including Random Forest, SVM, and XGBoost for classification.
- Integrated the trained model into a web based application for real time URL verification.

## EDUCATION

### Sphoorthy Engineering College

Bachelor of Technology, Computer Science & Engineering (AI & ML)

Hyderabad

Sep 2022 - Apr 2026

## CERTIFICATIONS

- AI Developer Intern: CYPWNG Software Technologies
- Python Programming Certificate: 2024